SOFTWARE REQUIREMENTS SPECIFICATION

Project 41

Doctor Prescription to Multilingual Video Reminder and Instruction Generator

Team Name: TechTitans

Team Details

Sl. No.	Member Name	SRN	Role
1	Asha	PES2UG24AM030	Quality, Security & Traceability Lead,
2	Chaya	PES2UG24AM045	Project Lead / Requirements Coordinator
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Date: 06/09/2026

Revision History

Version	Date	Description	Author
0.1	6/9/2026	Initial SRS structure prepared	TEAM 32
0.2	6/9/2026	Project introduction and overall description added	TEAM 32
0.3	6/9/2026	Team sections integrated and terminology reviewed	Project ID 41

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the requirements for Project 41 – Doctor Prescription to Multilingual Video Reminder and Instruction Generator.

The proposed system converts doctor-provided prescriptions into short multilingual video and audio medication reminders containing dosage instructions and timing alerts.

The system processes prescription information to identify medicine names, dosage, frequency, timing, and duration, and allows the extracted information to be verified before generating medication instructions and reminders.

The system supports multiple Indian languages so that patients can receive medication instructions in their preferred supported language.
The system is intended as an assistance and reminder tool and does not replace professional medical judgment.

1.2 Intended Audience

This SRS is intended for the following stakeholders:

- Doctors – to create, upload, review, and verify prescriptions.
- Patients – to receive multilingual medication instructions and reminders.
- System Administrators – to manage users and system configuration.
- Developers – to understand and implement the system requirements.
- Testers – to prepare test cases based on the specified requirements.
- Project Guide / Faculty – to review and evaluate the proposed system.

1.3 Product Scope

The proposed system provides a digital platform for converting doctor prescriptions into multilingual medication instructions and short video reminders.

The major functions within the scope are:

- Patient profile management.
- Prescription entry and upload.
- Prescription processing and information extraction.
- Medicine and dosage extraction.
- Prescription verification.
- Selection of a preferred Indian language.
- Translation of medication instructions.
- Audio and video reminder generation.
- Medication reminder scheduling.
- Medication tracking.
- Prototype pharmacy delivery/refill integration.

The system will not:

- Independently diagnose medical conditions.
- Independently prescribe medicines.
- Automatically modify the doctor's prescribed dosage.
- Replace professional medical advice.
- Generate final instructions from ambiguous prescription information without verification.
- Require a real commercial pharmacy API when one is unavailable for the academic prototype.

1.4 References

- Project 41 – Doctor Prescription to Multilingual Video Reminder and Instruction Generator.
- JACKFRUIT SRS Guideline – SRS structure and project requirements.
- Project 41 Team SRS Documentation.

2. OVERALL DESCRIPTION

2.1 Product Perspective

The Doctor Prescription to Multilingual Video Reminder and Instruction Generator is a software application that connects doctor-provided prescription information with patient medication reminders.

The system receives prescription information from an authorized user and processes it to obtain medicationrelated details. The extracted information is verified before being used to generate multilingual medication instructions and reminders.

Overall Workflow: Doctor → Prescription Upload/Entry → Prescription Processing/OCR → Medicine & Dosage Extraction → Verification → Language Selection → Translation → Audio/Video Generation →

Reminder Scheduling → Patient → Medication Tracking The major components of the system include:

- User and Patient Management
- Prescription Processing
- Medicine and Dosage Extraction
- Prescription Verification
- Multilingual Translation
- Text-to-Speech
- Video Generation
- Reminder Scheduling
- Medication Tracking
- Pharmacy Delivery Integration Database

External System Interaction

The system may use supporting services for:

- OCR-based prescription extraction
- Translation
- Text-to-Speech
- Video generation
- Notifications
- Pharmacy delivery/refill services

Key Product Characteristics

- Multilingual – supports multiple Indian languages.
- Patient-friendly – provides medication instructions through audio and video.
- Secure – protects patient and prescription information.
- Prescription-based – generates reminders using verified prescription information.
- Modular – major components can be developed and maintained independently.
- Scalable – additional Indian languages can be added in the future.

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Member 2 – Overall Description

Assigned Role: Requirements Analyst

2.2 Product Functions

This section provides a high-level summary of the major functions of the product. Detailed functional requirements are to be specified in the System Features section of the complete team SRS.

- Patient Management: Manage patient profiles and associate prescriptions and medication schedules with patients.
- Prescription Creation: Allow authorized users to enter or upload doctor prescriptions.
- Prescription Processing: Process prescription input and extract relevant prescription information.
- Medicine and Dosage Extraction: Identify medicine names, dosage information, frequency, timing, and durationfrom the prescription.
- Prescription Verification: Allow extracted information to be reviewed and corrected before reminders are generated.
- Language Selection: Allow the patient to select a preferred supported Indian language.
- Translation: Convert medication instructions into the patient's selected language.
- Video and Audio Reminder Generation: Generate short multilingual reminder content containing medicationinstructions.
- Reminder Scheduling: Schedule dosage and timing alerts according to the verified prescription.
- Medication Tracking: Maintain a basic record of scheduled, taken, or missed doses.
- Pharmacy Delivery Integration: Provide a prototype integration/interface for requesting medicine delivery or refillthrough pharmacy delivery services.

2.3 User Classes and Characteristics

The anticipated users of the system are categorized according to their responsibilities, access privileges, and expected interaction with the product.

2.3.1 Doctor

Role: Authorized user responsible for creating or verifying prescription information.

Functions used: Patient management, prescription creation/upload, review of extracted medicine and dosage information, and prescription verification.

Characteristics: Familiar with medical prescriptions and expected to have basic knowledge of the application.

2.3.2 Patient

Role: Primary recipient of multilingual medication instructions and reminders.

Functions used: View medication information, select preferred language, receive scheduled reminders, view generated video/audio instructions, and track medication doses.

Characteristics: May have varying levels of technical expertise and may prefer simple instructions in an Indian regional language.

2.3.3 System Administrator

Role: Authorized user responsible for administration and maintenance of the application.

Functions used: User management, system configuration, supported-language management, monitoring, and maintenance-related operations.

Characteristics: Expected to have appropriate technical knowledge and administrative privileges.

User Class	Primary Responsibilities
Doctor	Create/upload and verify prescriptions; manage prescription-related patient information.

Patient	Select language; receive medication instructions and reminders; track doses.
System Administrator	Manage users and system configuration; maintain the application.

2.4 Operating Environment

The system is intended to operate as a software application accessible through commonly used computing and mobile devices. The exact technology stack may be finalized by the project team during design and implementation.

2.4.1 Hardware Environment

- Desktop computers and laptops for doctors, administrators, and development/administration activities.
- Smartphones or tablets for patients and mobile access.
- Devices capable of displaying video and playing audio for reminder delivery.

2.4.2 Software Environment

- **Desktop/Web:** A modern web browser on Windows 10/11 or equivalent supported desktop operating systems. *(Proposed prototype environment; final versions may be fixed by the team.)*
- **Mobile:** A supported Android 10+ or iOS 15+ environment for patient access. *(Proposed prototype environment; final versions may be fixed by the team.)*
- A backend application server and database for processing and storing system information.
- OCR software/library or service for prescription text extraction.
- Translation software/service for multilingual medication instructions.
- Text-to-Speech software/service for audio generation.
- Video generation/avatar component for short reminder videos.

2.4.3 Network Environment

- Network/internet connectivity may be required for cloud-based OCR, translation, Text-to-Speech, video generation,notifications, and pharmacy delivery integration.
- The system should handle unavailable external services through suitable error handling.

2.4.4 Data Storage Environment

- The system may store patient profiles, prescription data, extracted medication information, reminder schedules,dosage logs, generated scripts/media, and pharmacy/refill request information.
- Access to stored prescription and patient information shall be restricted to authorized users.

2.5 Design and Implementation Constraints

2.5.1 Prescription/OCR Accuracy

Prescription images may be handwritten, unclear, or of varying quality. OCR results may therefore require verification and correction before they are used to generate medication instructions or reminders.

2.5.2 Multilingual Support

The system is required to support multiple Indian languages. The actual set of supported languages may depend on the availability and capability of translation and Text-to-Speech technologies.

2.5.3 Medical Information Integrity

The system must preserve important prescription information such as medicine, dosage, frequency, and timing. It should not independently alter a verified prescription.

2.5.4 External Services and APIs

OCR, translation, Text-to-Speech, video generation, notifications, and pharmacy delivery integration may depend on external services, APIs, libraries, or platform capabilities.

2.5.5 Pharmacy Delivery Integration

The official Project 41 scope includes pharmacy delivery integration. For the mini-project implementation, this may be represented through a prototype interface or mock integration where a real commercial API is not available.

2.5.6 Security and Privacy

Patient and prescription information is sensitive. The implementation must provide appropriate authentication, authorization, and protection against unauthorized access.

2.5.7 Platform and Resource Limitations

Video generation, media storage, API usage limits, processing time, and available computing resources may restrict the size, duration, or number of generated reminders.

2.5.8 Prototype Scope

The mini-project may use simplified or simulated components for functions that require commercial services or external integrations, while maintaining the intended system workflow.

2.6 Assumptions and Dependencies

2.6.1 Assumptions

- The doctor provides a valid prescription with sufficient information for processing.
- The uploaded prescription image is reasonably clear and contains readable medication information.
- Extracted prescription information is reviewed and verified by an authorized user before reminder generation.
- The patient provides accurate profile information and selects an appropriate supported language.
- Users have access to a compatible device capable of displaying the application's content.
- The patient can receive and play audio/video reminders on the selected device.
- Medication reminders are generated from the verified prescription information.
- Only languages supported by the implemented translation and Text-to-Speech components are available.
- Pharmacy delivery functionality may operate as a prototype/mock integration if a real pharmacy API is not available.
- The system is an assistance and reminder tool and does not replace professional medical judgment.

2.6.2Dependencies

Dependency	Purpose
OCR component/service	Extract text and medication information from prescription images.
Translation component/service	Translate medication instructions into supported Indian languages.
Text-to-Speech component/service	Convert reminder scripts into spoken audio.
Video/avatar component	Generate short video-based medication reminders.
Notification/scheduling mechanism	Deliver dosage and timing alerts at scheduled times.
Database	Store patient, prescription, medication, reminder, tracking, and integration-relate data.
Pharmacy delivery service/API	Support medicine delivery or refill requests; may be mocked in the prototype.
Network/internet connectivity	Enable access to external services and integrations where required.

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SOFTWARE REQUIREMENTS SPECIFICATION

Member 3 Contribution:

Suggested Role: System / Requirements Designer

3. EXTERNAL INTERFACE REQUIREMENTS

This section describes the user, software, and communications interfaces required by the system. It builds on the user classes (Doctor, Patient, System Administrator) and product functions defined in Section 2 of the team SRS.

3.1 User Interfaces

The system provides role based interfaces for the three identified user classes.

Doctor Interface

- Prescription Upload / Entry screen for submitting a prescription image or typed prescription details.
- Prescription Verification screen to review and correct extracted medicine, dosage, frequency, and timing information.
- Patient search and selection screen to associate a prescription with a registered patient.

Patient Interface

- Language Selection screen to choose a preferred supported Indian language.
- Reminder and Video Playback screen to view or replay generated medication instruction videos.
- Medication Tracking screen showing scheduled, taken, and missed doses.
- Notification and alert display for dosage and timing reminders.
- Pharmacy Delivery Request screen for requesting medicine delivery or refill.

System Administrator Interface

- User Management dashboard to create, update, or deactivate Doctor, Patient, and Administrator accounts.
- Supported Language configuration screen.
- System monitoring dashboard for tracking external service availability such as OCR, translation, Text to Speech, and video generation.

General UI Standards

- Interfaces shall use simple, uncluttered layouts suitable for users with varying levels of technical expertise.
- Patient facing screens shall be responsive and usable on smartphones and tablets.
- Interfaces shall support rendering of text in multiple Indian language scripts.
- Interfaces shall follow basic accessibility guidance such as adequate font size and color contrast.

3.2 Software Interfaces

The system depends on the following software components and services. Each interface is described in terms of its data input and output.

- OCR Component: Input is a prescription image in JPEG, PNG, or PDF format. Output is extracted text containing medicine name, dosage, frequency, timing, and duration.
- Translation Component: Input is verified prescription instructions in English. Output is translated instructions in the patient's selected Indian language.
- Text to Speech Component: Input is translated instruction text. Output is a generated audio file in the selected language.
- Video / Avatar Generation Component: Input is translated text and the associated audio file. Output is a short instructional video file.
- Database: Stores patient, doctor, administrator, prescription, medicine, dosage schedule, reminder, video, and pharmacy request records. Accessed by the backend application server.
- Pharmacy Delivery Service or Mock API: Input is a medicine or refill request. Output is a delivery or refill confirmation status.

3.3 Communications Interfaces

- The client application (web or mobile) communicates with the backend application server using HTTPS based REST APIs.
- Push notifications and or SMS are used to deliver dosage and timing reminder alerts to patients.
- Calls to external OCR, translation, Text to Speech, and video generation services are made over secure HTTPS connections.
- All communications carrying patient or prescription data shall be encrypted in transit to protect sensitive medical information.
- The system shall handle temporary unavailability of external services through appropriate error handling and retry mechanisms.

4. ANALYSIS MODELS

4.1 Use Case Diagram

The diagram below shows the three actors (Doctor, Patient, System Administrator) and the primary use cases they perform within the system.

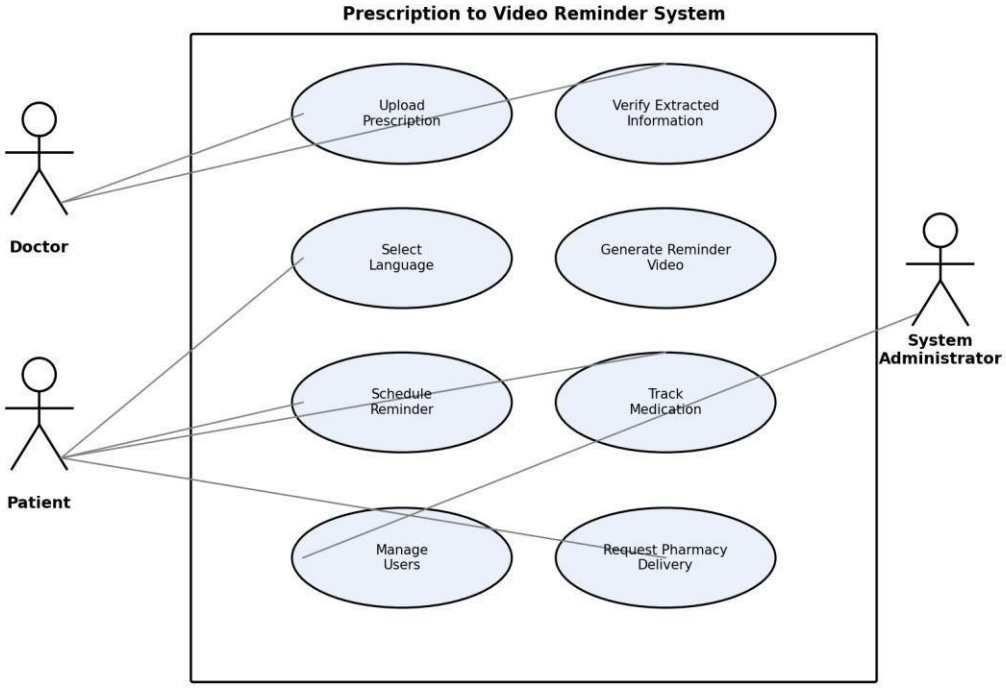


Figure 1: Use Case Diagram for Project 41

4.2 Use Case Descriptions

Detailed descriptions for each use case shown in Figure 1 are provided below.

Use Case ID	UC-01
Use Case Name	Upload Prescription
Actor	Doctor
Precondition	The Doctor is logged into the system and has selected a registered patient.
Main Flow	1. Doctor selects Upload Prescription. 2. Doctor provides a prescription image or enters prescription details manually. 3. System stores the prescription and forwards it for processing.
Postcondition	A new prescription record is created with status Pending Verification.
Exception Flow	If the image is unreadable or the upload fails, the system displays an error and asks the Doctor to retry.

Use Case ID	UC-02
Use Case Name	Verify Extracted Information
Actor	Doctor
Precondition	A prescription has been processed and medicine and dosage information has been extracted.
Main Flow	1. Doctor reviews the extracted medicine, dosage, frequency, and timing details. 2. Doctor corrects any inaccurate fields. 3. Doctor confirms the verified prescription.
Postcondition	The prescription status changes to Verified and becomes available for translation and reminder generation.

Exception Flow	If required fields are missing or invalid, the system prevents confirmation and highlights the fields needing correction.

Use Case ID	UC-03
Use Case Name	Select Language
Actor	Patient
Precondition	The Patient is registered and logged into the system.
Main Flow	1. Patient opens language settings. 2. Patient selects a preferred supported Indian language. 3. System saves the preference for future translation and video generation.
Postcondition	The patient's language preference is updated.
Exception Flow	If the selected language is not supported by the translation or Text to Speech component, the system informs the patient and lists available languages.

Use Case ID	UC-04
Use Case Name	Generate Reminder Video
Actor	System (triggered after verification)
Precondition	The prescription has been verified and the patient's preferred language is known.
Main Flow	1. System translates the verified instructions into the patient's language. 2. System generates audio using Text to Speech. 3. System generates a short instructional video using the audio and translated text.
Postcondition	A reminder video linked to the prescription is stored and made available to the patient.
Exception Flow	If translation, Text to Speech, or video generation fails, the system logs the error and retries or notifies the administrator.

Use Case ID	UC-05
Use Case Name	Schedule Reminder
Actor	Patient
Precondition	A verified prescription with dosage and timing information exists for the patient.
Main Flow	1. System reads the dosage schedule from the verified prescription. 2. System creates reminder entries for each scheduled dose. 3. System sends notifications at the scheduled times.
Postcondition	Reminder entries are created and notifications are queued for delivery.
Exception Flow	If the schedule conflicts with an existing reminder, the system flags the conflict for review.

Use Case ID	UC-06
Use Case Name	Track Medication
Actor	Patient
Precondition	At least one reminder has been sent to the patient.
Main Flow	1. Patient marks a dose as taken or missed. 2. System updates the medication adherence record.
Postcondition	The dosage log is updated with the current status.
Exception Flow	If the patient does not respond, the dose is marked as missed after the scheduled window closes.

Use Case ID	UC-07
Use Case Name	Manage Users

Actor	System Administrator
Precondition	The Administrator is logged into the system with administrative privileges.
Main Flow	1. Administrator searches for or adds a user account. 2. Administrator updates role, access, or status information. 3. System saves the changes.
Postcondition	The user account information is created or updated.
Exception Flow	If required fields are missing, the system rejects the change and displays a validation message.

Use Case ID	UC-08
Use Case Name	Request Pharmacy Delivery
Actor	Patient
Precondition	A verified prescription exists and the patient wishes to order or refill medicine.
Main Flow	1. Patient selects Request Delivery for a prescribed medicine. 2. System forwards the request to the pharmacy delivery service or mock integration. 3. System displays the confirmation status.
Postcondition	A pharmacy request record is created with a status such as Requested, Confirmed, or Failed.
Exception Flow	If the pharmacy service is unavailable, the system stores the request as Pending and notifies the patient.

4.3 ER Diagram

The entity relationship diagram below shows the main data entities used by the system and the relationships between them.

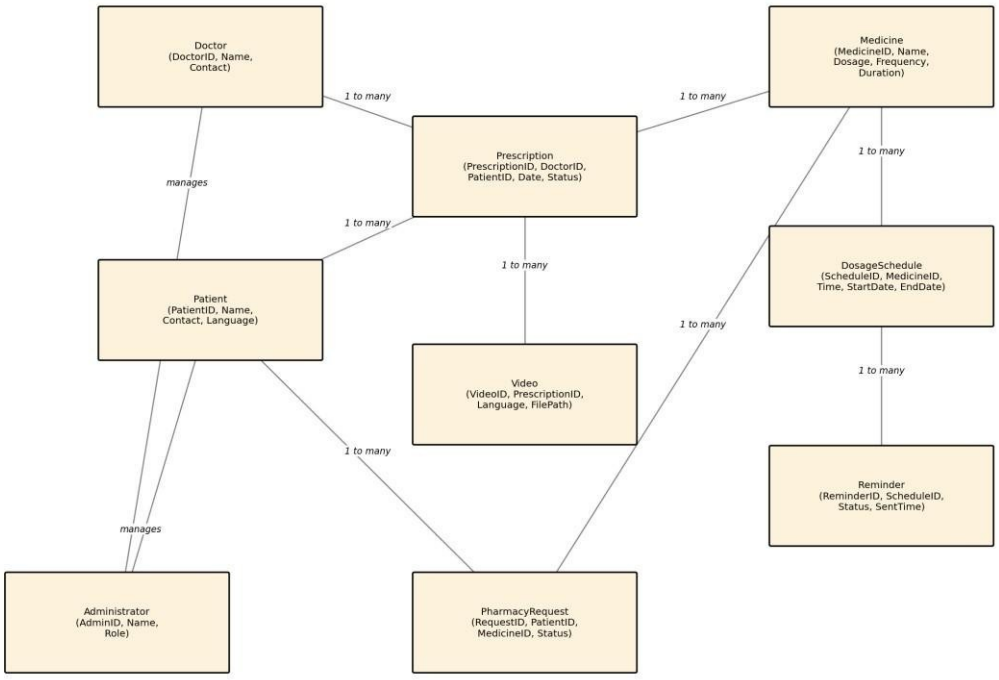


Figure 2: Entity Relationship Diagram for Project 41

5. SYSTEM FEATURES

This section describes the major system features in detail. Each feature includes a description and priority, a stimulus and response summary, and its associated functional requirements.

5.1 Prescription Management

Description and Priority: Allows storage and management of patient profiles and their associated prescriptions. Priority: High.

Stimulus / Response: When a Doctor uploads or enters a prescription for a registered patient, the system creates and stores a new prescription record linked to that patient.

Functional Requirements

- **REQ-PRES-01:** The system shall allow an authorized Doctor to create a prescription for a registered patient.
- **REQ-PRES-02:** The system shall validate mandatory prescription fields before saving.
- **REQ-PRES-03:** The system shall associate each prescription with exactly one patient and one prescribing doctor.
- **REQ-PRES-04:** The system shall maintain a prescription status such as Pending Verification, Verified, or Rejected.

5.2 Prescription Processing

Description and Priority: Extracts medicine, dosage, frequency, and timing information from an uploaded prescription using OCR and prepares it for Doctor verification. Priority: High.

Stimulus / Response: When a new prescription is uploaded, the system automatically runs OCR extraction and presents the extracted fields for review.

Functional Requirements

- **REQ-PROC-01:** The system shall extract medicine name, dosage, frequency, timing, and duration from an uploaded prescription image.
- **REQ-PROC-02:** The system shall present extracted information to the Doctor for review and correction before it is used further.
- **REQ-PROC-03:** The system shall not automatically finalize a prescription without Doctor verification.
- **REQ-PROC-04:** The system shall log OCR failures and allow manual entry as a fallback.

5.3 Multilingual Translation

Description and Priority: Converts verified medication instructions into the patient's selected Indian language. Priority: High.

Stimulus / Response: When a prescription is verified and a patient language preference is set, the system translates the instructions into that language.

Functional Requirements

- **REQ-TRANS-01:** The system shall translate verified medication instructions into the patient's selected supported language.
- **REQ-TRANS-02:** The system shall restrict language selection to the set of languages supported by the translation and Text to Speech components.
- **REQ-TRANS-03:** The system shall preserve medicine names, dosage values, and timing accurately during translation.

5.4 Video Generation

Description and Priority: Generates a short instructional video containing translated medication instructions. Priority: Medium.

Stimulus / Response: When translated instructions and generated audio are available, the system produces a short reminder video for the patient.

Functional Requirements

- **REQ-VID-01:** The system shall generate a short video containing translated medication instructions for each verified prescription.
- **REQ-VID-02:** The system shall store the generated video and link it to the corresponding prescription and patient.
- **REQ-VID-03:** The system shall allow the patient to replay a previously generated video at any time.

5.5 Reminder Management

Description and Priority: Schedules and delivers dosage and timing alerts based on the verified prescription. Priority: High.

Stimulus / Response: When a prescription is verified, the system creates a reminder schedule and sends notifications at the configured times.

Functional Requirements

- **REQ-REM-01:** The system shall generate a dosage reminder schedule from the verified prescription information.
- **REQ-REM-02:** The system shall send a notification to the patient at each scheduled dosage time.
- **REQ-REM-03:** The system shall allow the patient to mark a dose as taken or missed.
- **REQ-REM-04:** The system shall maintain a record of sent reminders and their corresponding dosage status.

SOFTWARE REQUIREMENTS SPECIFICATION

Member 4 Contribution

Quality, Security & Traceability Lead — Non-Functional Requirements, Other Requirements, Glossary, Field Layouts, Reports, and Requirement Traceability Matrix.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

The system shall provide timely responses while processing prescriptions, generating multilingual instructions, generating video reminders, and scheduling medication reminders. The following performance requirements shall be satisfied under normal operating conditions.

ID	Requirement
NFR-PERF-01	The system shall load the user dashboard within 3 seconds under normal operating conditions.
NFR-PERF-02	The system shall save a valid prescription within 2 seconds after submission, excluding delays caused by external services.
NFR-PERF-03	The system shall process a structured prescription and display the extracted medicine information within 5 seconds under normal operating conditions.
NFR-PERF-04	The system shall initiate multilingual instruction generation within 5 seconds after the user selects a supported language, excluding external service processing time.
NFR-PERF-05	The system shall schedule a valid medication reminder within 2 seconds after receiving all required reminder information.
NFR-PERF-06	The system shall display an appropriate processing or progress message when video generation requires additional processing time.
NFR-PERF-07	The system shall prevent accidental duplicate reminders for the same patient, medicine, date, and time.
NFR-PERF-08	The system shall remain responsive while long-running operations such as translation and video generation are being performed.

6.2 Safety Requirements

ID	Requirement
NFR-SEC-01	The system shall require authentication before allowing users to access protected functionality.

Since the system deals with doctor-provided prescription information, safety is a critical requirement. The system is intended to communicate and remind users about an existing prescription and shall not independently diagnose medical conditions or prescribe medicines.

ID	Requirement
NFR-SAFE-01	The system shall not independently prescribe medicines or recommend changes to a doctor's prescription.
NFR-SAFE-02	The system shall preserve the medicine name, dosage, frequency, and duration specified in the doctor's prescription.
NFR-SAFE-03	The system shall identify and report missing or invalid mandatory prescription information before processing.
NFR-SAFE-04	The system shall require verification of prescription information before generating final medication instructions.
NFR-SAFE-05	The system shall clearly indicate that generated instructions are based on the provided prescription and are not a substitute for professional medical advice.
NFR-SAFE-06	The system shall not silently modify medication names, dosage, frequency, or duration during processing or translation.
NFR-SAFE-07	If the system detects incomplete or ambiguous prescription information, it shall request verification instead of generating potentially incorrect instructions.
NFR-SAFE-08	The system shall allow an authorized user to correct prescription information before generating the final video reminder.

NFR-SAFE-09	The system shall display appropriate error messages when prescription processing fails.
NFR-SAFE-10	The system shall not generate a medication reminder when required prescription or scheduling information is invalid.

6.3 Security Requirements

The system handles patient and prescription information and therefore shall provide appropriate authentication, authorization, data protection, and secure communication mechanisms.

ID	Requirement
NFR-SEC-02	The system shall implement role-based access control for Doctor, Patient, and Administrator users.
NFR-SEC-03	The system shall restrict doctors to authorized patient and prescription records.
NFR-SEC-04	Patients shall only be permitted to access their own prescription, generated instructions, and reminder information.
NFR-SEC-05	User passwords shall not be stored in plain text.
NFR-SEC-06	The deployed system shall use HTTPS for communication between the client and server.
NFR-SEC-07	The system shall validate and sanitize user inputs to reduce security risks caused by malicious or invalid input.
NFR-SEC-08	The system shall prevent unauthorized users from accessing prescription records through manipulated requests.
NFR-SEC-09	The system shall implement session management to prevent unauthorized access through expired or inactive sessions.
NFR-SEC-10	The system shall maintain audit records for important operations such as prescription creation, modification, and deletion.
NFR-SEC-11	Sensitive patient and prescription information shall be protected against unauthorized access during storage and transmission.
NFR-SEC-12	Access to administrative functions shall be restricted to authenticated administrators.

6.4 Software Quality Attributes

6.4.1 Usability

ID	Requirement
NFR-QUAL-01	The system shall provide a simple and consistent user interface for doctors and patients.
NFR-QUAL-02	The system shall provide clear and meaningful error messages for invalid or incomplete inputs.
NFR-QUAL-03	The major functions of the system shall be understandable to users without requiring specialized technical knowledge.
NFR-QUAL-04	The system shall provide clear navigation between prescription, instruction, video, and reminder functionalities.

6.4.2 Reliability

ID	Requirement
NFR-QUAL-05	The system shall preserve successfully stored prescription information during normal application failures.
NFR-QUAL-06	The system shall handle unavailable translation, text-to-speech, video-generation, or notification services gracefully.

NFR-QUAL-07	The system shall display an appropriate failure message when an operation is unsuccessful instead of displaying a false success message.

6.4.3 Maintainability

ID	Requirement
NFR-QUAL-08	The system shall use modular components so that prescription processing, translation, video generation, and reminder management can be modified independently.
NFR-QUAL-09	The source code shall follow agreed coding standards and include appropriate documentation.
NFR-QUAL-10	Changes to one major module should have minimal impact on other independent modules.

6.4.4 Testability

ID	Requirement
NFR-QUAL-11	Major system modules shall have testable interfaces for unit and integration testing.
NFR-QUAL-12	Functional and nonfunctional requirements shall be mapped to corresponding test cases through the Requirement Traceability Matrix.

6.4.5 Scalability

ID	Requirement
NFR-QUAL-13	The architecture shall allow additional Indian languages to be added without major modification to existing prescription-processing functionality.
NFR-QUAL-14	The reminder component shall support multiple medicines and multiple scheduled reminders for a patient.

6.4.6 Portability

ID	Requirement
NFR-QUAL-15	The application shall be deployable using Docker to provide a consistent execution environment.

6.4.7 Availability

ID	Requirement
NFR-QUAL-16	The system shall be available to authorized users whenever the application services are operational.
NFR-QUAL-17	Temporary failure of an external service shall not cause permanent loss of prescription information.

6.5 Business Rules and Domain Requirements

ID	Requirement
BR-01	Only an authenticated and authorized doctor shall be permitted to create or modify a patient's prescription.
BR-02	A patient shall only be permitted to view their own prescription and reminder information.
BR-03	The system shall use the medication information supplied in the doctor's prescription as the source for generating instructions.
BR-04	The system shall not automatically modify a doctor's prescribed dosage.
BR-05	A reminder shall require valid medicine, frequency, and scheduling information.

BR-06	Only languages supported by the system shall be available for instruction generation.
BR-07	A generated video shall correspond to the selected patient, prescription, medicine, and language.
BR-08	Invalid or cancelled prescriptions shall not generate new medication reminders.
BR-09	If prescription information is incomplete or ambiguous, the system shall request verification before generating instructions.
BR-10	The system shall maintain records of important prescription and reminder activities for accountability and troubleshooting.
BR-11	The system shall treat the doctor's prescription as the authoritative source of medication information.
BR-12	The system shall not provide medical diagnosis or independently recommend medication.

7. Other Requirements 7.1

Database Requirements

ID	Requirement
OR-DB-01	The system shall maintain a unique identifier for each registered user.
OR-DB-02	Each prescription shall have a unique prescription identifier.
OR-DB-03	Each prescription shall be associated with the corresponding patient and authorized doctor.
OR-DB-04	Each reminder shall be associated with the corresponding patient, prescription, medicine, and scheduled time.
OR-DB-05	The database shall maintain appropriate relationships between users, patients, prescriptions, medicines, reminders, and generated videos.
OR-DB-06	The system shall validate mandatory database fields before storing records.
OR-DB-07	The system shall prevent unauthorized modification or deletion of stored prescription information.
ID	**Requirement**
OR-DB-08	The system shall support backup and recovery procedures for important application data in the deployed environment.

7.2 Internationalization and Multilingual Requirements

Multilingual support is a major objective of Project 41. The system shall support multiple Indian languages for communicating prescription instructions and reminders.

ID	Requirement
OR-I18N-01	The system shall support multiple Indian languages selected by the user.
OR-I18N-02	The selected language shall be applied consistently to generated medication instructions.
OR-I18N-03	The system shall use Unicode-compatible data storage and processing for supported languages.
OR-I18N-04	The system shall allow additional supported languages to be added without redesigning the entire system.

OR-I18N-05	If a requested language is unavailable, the system shall clearly notify the user and display the available language options.
OR-I18N-06	Translation shall preserve important medication information such as medicine name, dosage, frequency, and duration.
OR-I18N-07	Generated multilingual content shall be checked for missing or incorrectly translated information before final output.

7.3 Privacy Requirements

ID	Requirement
OR-PRIV-01	The system shall collect only information required to provide its intended functionality.
OR-PRIV-02	Patient and prescription information shall not be displayed to unauthorized users.
OR-PRIV-03	The system shall implement access controls for stored patient information.
OR-PRIV-04	Generated videos containing patient-specific information shall only be accessible to authorized users.
OR-PRIV-05	The system shall follow the project's defined data retention and deletion policy.
OR-PRIV-06	Patient information shall not be unnecessarily shared with external services.
OR-PRIV-07	Data transmitted to external services, where required, shall be limited to the information necessary for the requested operation.

7.4 Accessibility Requirements

ID	Requirement
OR-ACC-01	The system shall use readable text and clearly identifiable interface controls.
OR-ACC-02	Important information shall not be communicated using colour alone.
OR-ACC-03	Video instructions should contain readable on-screen text along with audio where applicable.
OR-ACC-04	The system shall provide clear and understandable error messages.
OR-ACC-05	The major web functions should support keyboard-based navigation where feasible.
OR-ACC-06	The system shall use simple and understandable language for patient-facing instructions.

7.5 Legal and Domain Considerations

The system is intended as an academic software prototype for communicating and reminding users about information contained in a doctor-provided prescription.

ID	Requirement
OR-LEGAL-01	The system shall clearly communicate that it does not independently diagnose medical conditions or prescribe medicines.
OR-LEGAL-02	The system shall preserve the original prescription information provided by an authorized user.

ID	Requirement

OR-LEGAL-03	Patient information shall be handled according to applicable privacy and data-protection requirements.
OR-LEGAL-04	Third-party translation, text-to-speech, notification, and video-generation services shall be used according to their applicable terms and data-handling policies.
OR-LEGAL-05	The system shall not claim medical certification unless the required certification has actually been obtained.
OR-LEGAL-06	The project team shall consider applicable healthcare, privacy, accessibility, and software regulations before deployment beyond the academic prototype.

Appendix A: Glossary

Term	Definition
API	Application Programming Interface used for communication between software components.
Authentication	Process of verifying the identity of a user.
Authorization	Process of determining what an authenticated user is permitted to access.
Dosage	The quantity of medicine that should be taken at a specified time.
Frequency	The number of times or schedule at which a medicine should be taken.
Multilingual	Supporting more than one language.
NFR	Non-Functional Requirement describing qualities or constraints of the system.
Prescription	Medication instructions provided by an authorized doctor.
RBAC	Role-Based Access Control used to restrict functionality according to user roles.
Reminder	A scheduled notification informing a patient about medication timing.
SRS	Software Requirements Specification.
TTS	Text-to-Speech technology used to convert written text into spoken audio.
Translation	Conversion of content from one language to another.
Video Reminder	A short video containing medication instructions and reminder information.
Audit Log	A record of important activities performed within the system.
Unicode	A standard used to represent text from different languages and writing systems.
Traceability Matrix	A document that maps requirements to architecture, design, implementation, and testing artifacts.
Functional Requirement	A requirement describing a function or behaviour that the system must provide.
Patient	A user who receives prescription information and medication reminders.
Doctor	An authorized user who creates or manages patient prescriptions.

Appendix B: Field Layouts

B.1 Patient Registration

Field	Length	Data Type	Description	Is Mandatory
Patient ID	20	Alphanumeric	Unique identifier of the patient	Y
Patient Name	100	String	Full name of the patient	Y
Date of Birth	10	Date	Date of birth of the patient	Y

Phone Number	15	String	Contact number of the patient	Y
Email	100	String	Email address of the patient	N
Preferred Language	30	String	Language preferred for instructions	Y
Address	250	String	Address of the patient	N
Status	20	String	Active or inactive status	Y

B.2 Prescription

Field	Length	Data Type	Description	Is Mandatory
Prescription ID	20	Alphanumeric	Unique prescription identifier	Y
Patient ID	20	Alphanumeric	Identifier of the associated patient	Y
Doctor ID	20	Alphanumeric	Identifier of the doctor	Y
Medicine Name	100	String	Name of the prescribed medicine	Y
Dosage	50	String	Prescribed dosage	Y
Frequency	50	String	Frequency of medication	Y
Duration	30	String	Duration of treatment	Y
Instructions	500	String	Additional instructions provided by doctor	N
Prescription Date	10	Date	Date on which prescription was created	Y
Status	20	String	Active, completed, or cancelled	Y

B.3 Reminder

Field	Length	Data Type	Description	Is Mandatory
Reminder ID	20	Alphanumeric	Unique reminder identifier	Y
Patient ID	20	Alphanumeric	Identifier of the patient	Y
Prescription ID	20	Alphanumeric	Associated prescription	Y
Medicine Name	100	String	Medicine associated with the reminder	Y
Reminder Time	8	Time	Scheduled time of reminder	Y
Start Date	10	Date	Date from which reminder starts	Y
End Date	10	Date	Date on which reminder ends	Y
Status	20	String	Active, completed, or disabled	Y

B.4 Generated Video

Field	Length	Data Type	Description	Is Mandatory
Video ID	20	Alphanumeric	Unique video identifier	Y
Prescription ID	20	Alphanumeric	Associated prescription	Y
Patient ID	20	Alphanumeric	Associated patient	Y
Language	30	String	Language used in the video	Y
Field	**Length**	**Data Type**	**Description**	**Is Mandatory**
Video Path/URL	500	String	Location of generated video	Y
Generation Status	20	String	Processing, completed, or failed	Y
Created Date	20	DateTime	Date and time of video creation	Y

B.5 Reminder/Medication Status

Field	Length	Data Type	Description	Is Mandatory
Status ID	20	Alphanumeric	Unique status record identifier	Y
Reminder ID	20	Alphanumeric	Associated reminder	Y
Patient ID	20	Alphanumeric	Associated patient	Y
Scheduled Time	20	DateTime	Scheduled medication time	Y
Action Status	20	String	Taken, missed, or pending	Y
Recorded Time	20	DateTime	Time at which status was recorded	N

B.6 Prescription Report

The Prescription Report shall contain: Prescription ID, Patient ID, Patient Name, Doctor ID, Medicine Name, Dosage, Frequency, Duration, Prescription Date, and Status.

B.7 Medication Reminder Report

The Medication Reminder Report shall contain: Reminder ID, Patient Name, Medicine Name, Dosage, Scheduled Time, Start Date, End Date, and Reminder Status.

B.8 Video Generation Report

The Video Generation Report shall contain: Video ID, Patient ID, Prescription ID, Language, Generation Status, and Created Date.

B.9 Medication Adherence Report

The Medication Adherence Report shall contain: Patient ID, Medicine Name, Scheduled Date and Time, Reminder Status, Taken/Missed Status, and Recorded Date and Time.

Appendix C: Requirement Traceability Matrix

At the initial SRS stage, architecture, design, code, and test references may not yet be finalized. Therefore, TBD is used where the corresponding artifact will be created during later phases. The matrix shall be updated throughout the SDLC.

Sl. No.	Requirement ID	Brief Description	Architecture	Design	Code	Test Case	System Test

1	NFR-PERF-01	Dashboard shall load within 3 seconds	Presentation Layer	Dashboard Design	TBD	TBD	TBD
2	NFR-PERF-05	Reminder shall be scheduled within 2 seconds	Reminder Module	Reminder Design	TBD	TBD	TBD
3	NFR-SAFE-01	System shall not prescribe medicines	Application Layer	Safety Rules	TBD	TBD	TBD
4	NFR-SAFE-02	Prescription dosage shall be preserved	Prescription Module	Prescription Design	TBD	TBD	TBD
5	NFR-SEC-01	Authentication shall be required	Authentication Module	Login Design	TBD	TBD	TBD
6	NFR-SEC-02	Role-based access shall be implemented	Security Module	RBAC Design	TBD	TBD	TBD
7	NFR-SEC-04	Patients shall access only their own data	Security Module	Access Control Design	TBD	TBD	TBD
8	NFR-QUAL-08	System shall use modular components	Application Layer	Component Diagram	TBD	TBD	TBD
9	NFR-QUAL-13	System shall support additional languages	Translation Module	Translation Design	TBD	TBD	TBD
10	OR-DB-02	Each prescription shall have a unique ID	Database Layer	ER Diagram	TBD	TBD	TBD
11	OR-I18N-01	System shall support multiple Indian languages	Translation Module	Multilingual Design	TBD	TBD	TBD
12	OR-I18N-03	System shall support Unicode	Translation Module	Data Design	TBD	TBD	TBD
13	OR-PRIV-02	Unauthorized users shall not access patient data	Security Module	Access Control Design	TBD	TBD	TBD
14	OR-ACC-03	Video shall contain readable text	Video Module	Video Design	TBD	TBD	TBD
15	OR-LEGAL-01	System shall not independently diagnose or prescribe	Application Layer	Domain Rules	TBD	TBD	TBD

Traceability Flow: Requirement → Architecture → Design → Code → Unit Test → System Test